

ME6404 – Fall 2025

Lab 2 – Input Shaping

PART 1. INPUT SHAPING ON THE BRIDGE CRANE

SAFETY

WARNING: This robotic apparatus is inherently dangerous and includes an enable/disable feature. Before any student enters the robot's operating zone, the system should be either disabled or fully powered off.

OBJECTIVES

- Use input shaping to move a suspended payload in a point-to-point motion with low vibration.
- Compare the effect of move distance on shaped and unshaped commands.
- Compare the effect of changes in system frequency on various input shapers.

BACKGROUND

You will demonstrate the benefits of input shaping for point-to-point moves. You will also investigate how input shaping is affected by variations in move distance and system frequency.

In the previous lab, you modified a trapezoidal velocity profile and then held down the move button to execute the profile. Figure 1 shows what the controller is doing. Pressing the button at time t_A and holding it down until time t_B will generate a step input of duration $t_B - t_A$. This is then convolved with the impulse sequences that you designed to move the trolley a specified distance. However, due to the inertia of the crane and motor limitations, the actual velocity profile resembles a trapezoid. Note that the negative impulse at t_f is used to bring the crane to a stop despite you continuing to hold down the move button. Note that the impulses sum to zero, rather than one. Therefore, this is not a standard input shaper procedure. Rather, it is a modified version that always moves the crane the same distance and will return the trolley to the initial position when the button is released. Varying t_f will change the length of the velocity profile and, therefore, the total move distance.

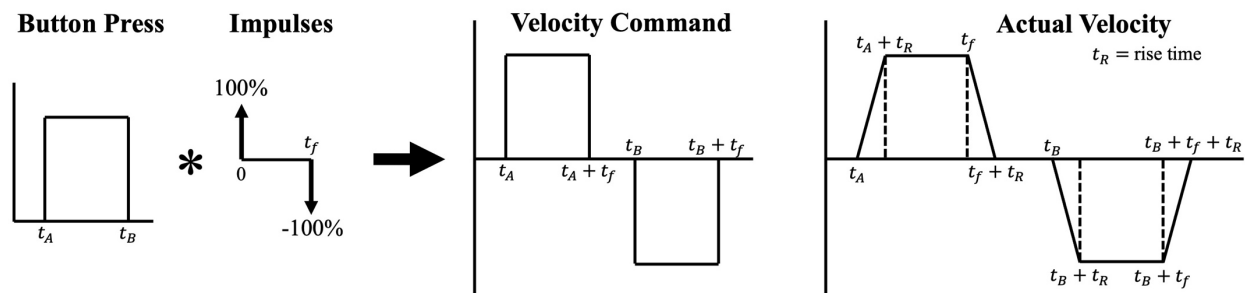


Figure 1: Convolution using the “Pulse” velocity profile.

The impulse sequence in Figure 1 is used to move the trolley the desired distance and can be convolved with a ZV shaper, as seen in Figure 2. Note that the last two impulses of the convolved sequence are like the first two, except they are negative in amplitude.

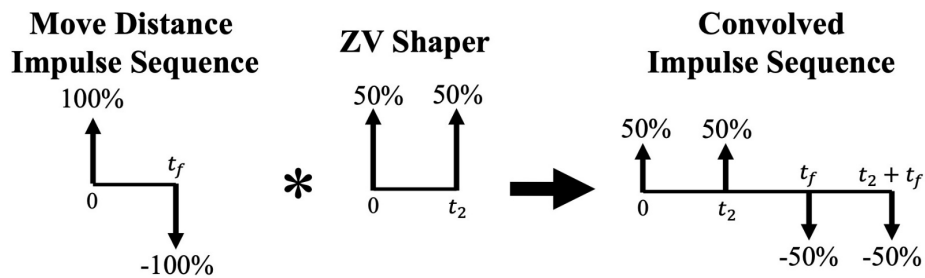


Figure 2: Convolution of move distance impulse sequence with a ZV shaper.

Figures 3a and 3b show the process of moving the trolley with a ZV input shaper.

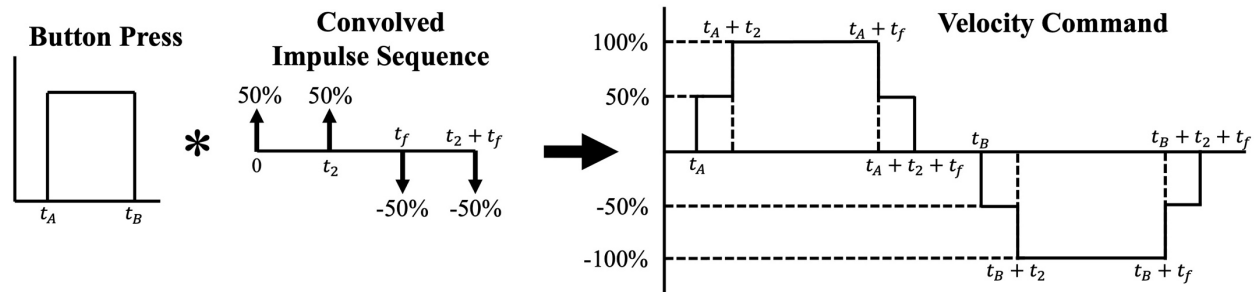


Figure 3a: Convolution using “ZV Shaper Return”.

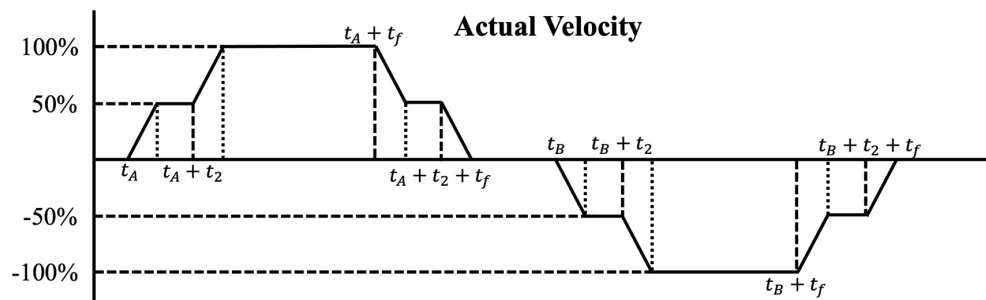


Figure 3b: Actual velocity using “ZV Shaper Return”.

LAB TASKS

Part A: Variation in Move Distance (Use the Trolley Axis)

1. Move the crane using the trapezoidal velocity profile with 100% speed. Use a 0.8 m cable length (measured from the center of the hoist drum to the center of the payload). Do this for $t_f = 1100\text{ms}$, 1600ms , and 2000ms . Measure the residual payload oscillations.
 - Record at least 4-5 payload oscillations.
2. Repeat task A1 using a ZV shaper of your own design that you think will best minimize the payload oscillations.

Part B: Variation in System Frequency

1. Move the crane using a trapezoidal velocity profile at 100% speed, with $t_f = 1500\text{ms}$. Do this for cable lengths of 0.6 m, 0.9 m, and 1.2 m. Measure the residual payload oscillations.
 - Record at least 4-5 oscillations.

2. Repeat task B1 using only one ZV shaper, for all three cable lengths, that you think will best minimize the oscillations for all cable lengths.
3. Repeat task B1 using only one robust shaper (SI, EI, ZVD, ZVDD, etc.) of your own design.

LAB PROCEDURE

1. Sign up for sessions on the bridge crane Outlook calendar.
2. Figure 4 shows the shaper design interface of the GUI.
3. Use the **Amplitude** and **Time** fields to modify the impulse amplitudes and times. Always hit Enter on the keyboard after you modify a text box. The first impulse should occur at time = 0. To bring the crane to a stop, the last n impulses should mirror the first n impulses, but with negative amplitudes.
 - Note: the input shaper you are using is only the 1st n impulses you enter into the table. The 2nd n impulses arise from the step down at the end of the velocity pulse.
4. Return to the main window using the **back** button.
5. Record data with the same procedure as Lab 1A.

Input shaper edit form					
	Amplitude	Time (ms)		Amplitude	Time (ms)
Impulse 1:	50	0	Imp 11:	0	0
Impulse 2:	50	500	Imp 12:	0	0
Impulse 3:	-50	2000	Imp 13:	0	0
Impulse 4:	-50	2500	Imp 14:	0	0
Impulse 5:	0	0	Imp 15:	0	0
Impulse 6:	0	0	Imp 16:	0	0
Impulse 7:	0	0	Imp 17:	0	0
Impulse 8:	0	0	Imp 18:	0	0
Impulse 9:	0	0	Imp 19:	0	0
Impulse 10:	0	0	Imp 20:	0	0

BACK

Figure 4: Bridge crane shaper design interface.

PART 2. INPUT SHAPING ON TOWER CRANE

SAFETY

WARNING: This robotic apparatus is inherently dangerous and includes an enable/disable feature. Before any student enters the robot's operating zone, the system should be either disabled or fully powered off.

OBJECTIVES

- Investigate input shaping for multi-mode systems and rotational motion.

BACKGROUND

On the tower crane in Lab 1, you specified the magnitude and duration of the step profile you wanted to apply to the system. However, due to the inertia of the crane and motor limitations, the actual velocity profile resembles a trapezoid, as shown in Figure 5. Because you can specify the duration of the step profile directly on the tower crane, this specified step profile can be directly convolved with a shaper. The command and actual trolley velocities after the step profile is convolved with a ZV shaper are shown in Figure 6.

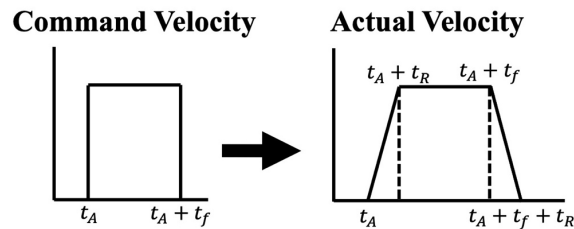


Figure 5: Command and actual velocity of crane trolley for specified step input.

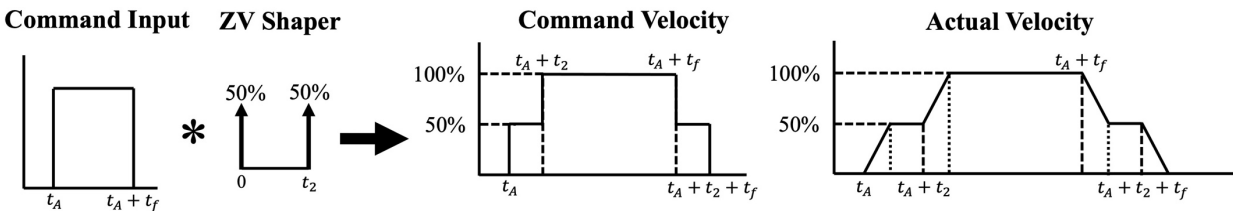


Figure 6: Convolution of specified step input with a ZV shaper.

Certain crane payload configurations can result in a double-pendulum with two modes of oscillation, as illustrated in Figure 7. These two-mode oscillations are nearly impossible to eliminate by a human operator without assistance from an auxiliary controller.

In Figure 7, the hook suspension length and the payload rigging lengths are L_1 and L_2 , respectively. The mass of the hook and payload are m_h and m_p , respectively.

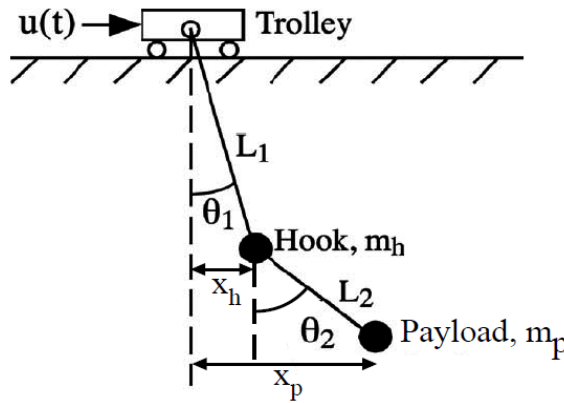


Figure 7: Double-pendulum model.

The linearized natural frequencies of this system are given by (assuming zero damping):

$$\omega_{1,2} = \sqrt{\frac{g}{2} \left((1+R) \left(\frac{1}{L_1} + \frac{1}{L_2} \right) \mp \beta \right)}$$

$$R = \frac{m_p}{m_h}$$

$$\beta = \sqrt{(1+R)^2 \left(\frac{1}{L_1} + \frac{1}{L_2} \right)^2 - 4 \left(\frac{1+R}{L_1 L_2} \right)}$$

The parameters for this lab are shown in Table 1.

Table 1 - Hook and Payload Information

Hook mass, m_h	0.407 kg
Payload mass, m_p	0.569 kg
L_1	1.000 m
L_2 (eye of hook to payload center)	0.200 m

LAB TASKS

1. The suspension length (L_1) (**Hoist Length**) should be set to roughly 0.6 m (within 2 or 3 mm). Then **slew** the crane (use CW/CCW buttons) using a pulse velocity profile at 100% speed (see step 5 in the Lab Procedure below) and $t_f = 4000\text{ms}$ (**start from a slew angle of $\sim 80^\circ$**). Do this for three different radial trolley positions on the jib (use the In/Out buttons): “**Trolley Position**” = **500mm**, **700mm**, and **900mm**. Record the tangential and radial residual hook oscillations (4-5 oscillations).
2. Repeat task 1 with the same t_f , the same Hoist Pos, the same three Trolley Positions, and use **only one single-mode ZV shaper** of your own design that you think will best minimize the hook (and payload) oscillations.
3. Repeat task 1 using **only one single-mode robust shaper** (e.g., SI, EI, ZVD, ZVDD).
4. Repeat task 1 using **only one two-mode shaper**.

LAB PROCEDURE

1. Sign up for sessions using the tower crane Outlook calendar.
2. Follow step 1 of Part 1B in “Lab 1 Bridge&Tower 2025” to power up the tower crane system.
3. Open the tower crane GUI.
 - a. Log in to the laptop using the password **CraneSafety!2025A**
 - b. Launch the tower crane GUI via the desktop shortcut shown in Figure 8.
 - c. This will open a terminal window. Ctrl+Click the link shown in Figure 9 to open the GUI in a web browser.



Figure 8: Location of shortcut for tower GUI on the desktop.

```
Crane GUI
Loaded calibration from file: 24.01 471.98 902.8
Connecting to COM7 at 74880 baud.
Connected to COM7.
Calibration command sent on connect: c,24.01,471.98,902.8
Crane Status: Starting ODriveCAN
Crane Status: Waiting for ODrives...
Crane Status: Found Slewing
Crane Status: Found Trolley
Crane Status: Found Hoisting
Crane Status: Motor driver feedback setup, blocking for homing.
Crane Status: Homing complete, setting to closed loop control.
Crane Status: ODrive running!
Crane Status: Calibrating -1.00,-1.00,902.80
Camera connected.
* Local: http://127.0.0.1:5000
* Serving Flask app 'app'
* Debug mode: off
```

Figure 9: Tower crane GUI launch terminal window.

d. The interface for the tower crane is shown in Figure 10. Wait for the page to finish loading.

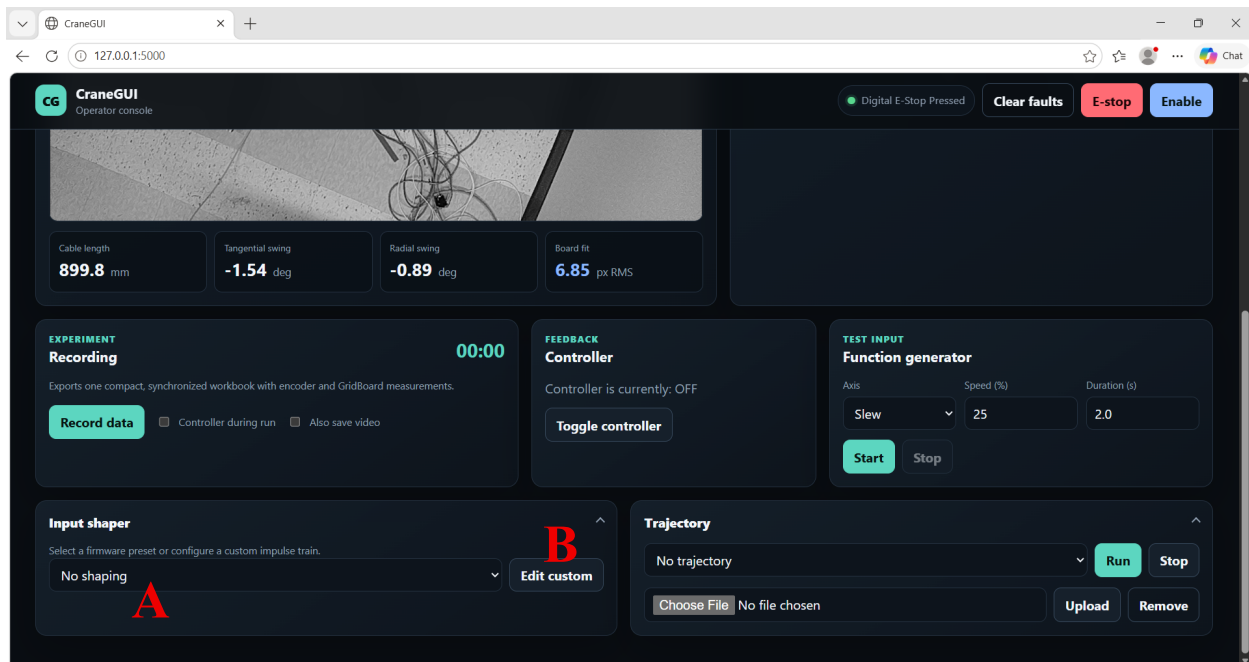


Figure 10: Tower crane GUI interface.

- e. Make sure the camera stream in the GUI is running as seen in Figure 10.
 - i. If the camera stream is not running, close the browser and reopen it. Be patient; the camera stream can take several seconds to spin up.
4. Refer to step 3 of part 1B in “Lab 1 Bridge&Tower 2025” for directions on how to use the Function Generator and collect the unshaped data specified in the Lab Tasks section.

5. The Input Shaper card has a dropdown indicated by **A** in Figure 10 that allows the user to select which shaper to use. Note that you should not be using any of the premade input shapers for this specific lab, only your own shapers. To add your custom shapers, first press the Edit Custom Shaper button indicated by **B** in Figure 10 to pull up the window shown in Figure 11.
6. To input your shaper, use the following procedure (all letters are in reference to Figure 11).
 - a. Select the correct number of impulses using either the input box on the left marked with **C** or the row increment/decrement buttons marked with **D**.
 - b. Input your impulse amplitudes into the column marked **E** and your delays into the column marked **F**.
 - c. Press the Apply & Activate button marked with **G**.
 - d. Close the custom shaper window to go back to the main GUI using the button marked **H**.
7. Input your shapers using the instructions in 6 and conduct the remaining tests specified in the Lab Tasks section of Part 2 of this lab handout.

Figure 11: Tower crane GUI custom shaper interface.

Lab Writeup

Present your work using **less than 3 pages of text** (not including Figures). Your report should include the following:

Portable Bridge Crane:

Part A:

Compare the residual oscillations for all trials. Explain any trends in the data and make reasonable conclusions. You must justify and explain your design for the ZV shaper. Include appropriate and relevant plots.

Part B:

Compare the residual oscillations for all trials. Explain any trends in the data and make reasonable conclusions. You must justify and explain your selection and design of the ZV and robust shapers. Include appropriate and relevant plots.

Mobile Tower Crane:

Compare the residual oscillations for all trials. In addition, address the following questions: How does the slewing motion affect residual oscillations? Does the radial distance of the trolley on the jib have any effect? How did the slewing motion (as opposed to translational) affect the design of your shaper? **Select the best shaper** for this application and explain why it is the best. Use experimental data that you collected and consider the various tradeoffs in your discussion. Include appropriate and relevant plots/images.

You should think carefully about the plots that you include in your report – these must be neat, clear, non-redundant, and efficiently convey the desired message. Furthermore, in the text, you should comment appropriately on the plots (which must be appropriately placed in the text) and discuss what is significant about the results shown.